

MLCC 2026
Regularization Networks II: beyond linear models

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About this class

- ▶ Extend our model to deal with non linear problems
- ▶ Formulate the Representer Theorem
- ▶ Introduce kernel functions (+ examples)

Linear model...

- ▶ Data set $S = \{(x_1, y_1), \dots, (x_n, y_n)\}$ with $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$
- ▶ $\hat{X} = (x_1, \dots, x_n)^\top \in \mathbb{R}^{n \times d}$ and $\hat{Y} = (y_1, \dots, y_n)^\top$.
- ▶ Linear model $w \in \mathbb{R}^d$: $y \approx f_w(x) = w^\top x$
- ▶ Tikhonov regularization

Summary: our *optimal* regression function will be f_{w^*} , where

$$w_\lambda = \arg \min_{w \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_w(x_i)) + \lambda \|w\|^2.$$

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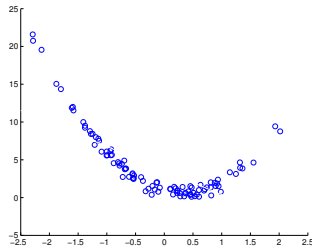
Recall: $w_\lambda = (\hat{X}^\top \hat{X} + \lambda n I)^{-1} \hat{X}^\top \hat{Y}$.

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Example $d = 1$ and S as in the plot.



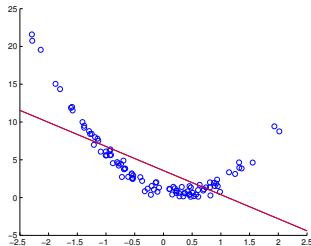
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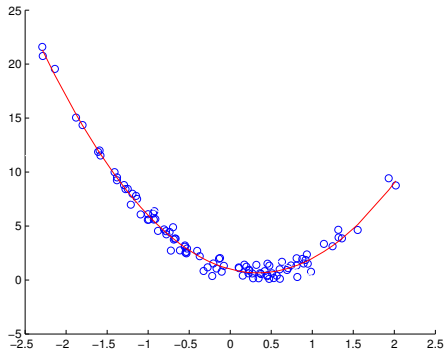
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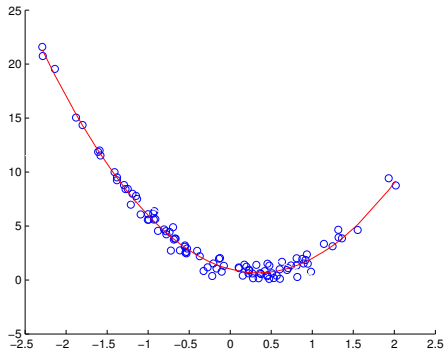
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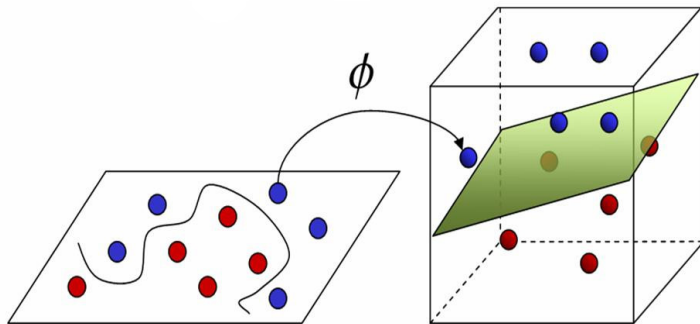
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Geometric view

$$f(x) = w^\top \Phi(x)$$



Input Space

Feature Space

Non linear models

- ▶ Let $\varphi_j(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ with $j \in \{1, \dots, D\}$ (in general with $D \gg d$)
- ▶ $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^D$ is called *feature map* with $\phi(x) = (\varphi_1(x), \dots, \varphi_D(x))^\top$.
- ▶ $w \in \mathbb{R}^D$.

Nonlinear model

$$f_w(x) = w^\top \phi(x) = \sum_{j=1}^D w_j \varphi_j(x)$$

How to compute a non linear model (least squares)

Let $\hat{\Phi} = (\phi(x_1), \dots, \phi(x_n))^{\top} \in \mathbb{R}^{n \times D}$.

$\hat{\Phi}$ is the data matrix in the feature space (simply $\hat{X}$ if ϕ is the identity).

For Regularized Least Squares the explicit minimizer w for the Empirical Loss is (same calculation because the problem is still linear in w)

$$w = (\hat{\Phi}^{\top} \hat{\Phi} + \lambda n I)^{-1} \hat{\Phi}^{\top} \hat{y}$$

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(from a computational point of view)

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Representer Theorem (in the least squares context)

If w solves RLS then

$$w = \hat{\Phi}^\top c = \sum_{i=1}^n c_i \phi(x_i),$$

where $c = (\hat{\Phi} \hat{\Phi}^\top + \lambda n I_n)^{-1} \hat{Y} \in \mathbb{R}^n$ and $\hat{\Phi} \hat{\Phi}^\top \in \mathbb{R}^{n \times n}$.

Sketch of the Proof

We want to show that $\omega = (\hat{\Phi}^\top \hat{\Phi} + \lambda n I_D)^{-1} \hat{\Phi}^\top \hat{Y} = \hat{\Phi}^\top (\hat{\Phi} \hat{\Phi}^\top + \lambda n I_n)^{-1} \hat{Y}$. Is it true that

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Yes, it is true (in general $(BA + \lambda Id)^{-1} B = B(AB + \lambda Id)^{-1}$, whenever A and B are compatible)!

Generalization of Representer Theorem for any Loss Functions

For a given loss function $\ell : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, let the problem be

$$w^* = \arg \min_{w \in \mathbb{R}^D} \frac{1}{n} \sum_{i=1}^n \ell(y_i, \phi(x_i)^\top w) + \lambda \|w\|^2$$

The solution can always be written as $w^* = \hat{\Phi}^\top c$ for some coefficients vector $c = (c_1, \dots, c_n)$

Representer Theorem for general Loss Functions

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$$w = \hat{w} + w_\perp \quad \text{for each } w \in \mathbb{R}^D$$

with $\hat{w} \in \hat{W}$ and $v^\top w_\perp = 0$ for each $v \in \hat{W}$.

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That is

$$w^* = \hat{\Phi}^\top c$$

for some $c \in \mathbb{R}^n$.

Why we need Kernels...

Let us analyze the regression function f_w , which minimizes RLS for the Generalized Linear model:

$$f_w(x) = \phi(x)^\top \omega = \phi(x)^\top \hat{\Phi}^\top c = \sum_{i=1}^n \phi(x)^\top \phi(x_i) c_i$$

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$$c = (\hat{\Phi} \hat{\Phi}^\top + \lambda n I)^{-1} \hat{Y}$$

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$f_w(x)$ **is expressed only by using inner products between feature vectors**

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In this way we have

$$f_w(x) = \hat{K}_x^\top (\hat{K} + \lambda n I_n)^{-1} \hat{Y}$$

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The same holds for general loss functions indeed

$$f_{w^*}(x) = \phi(x)^\top w^* = \phi(x)^\top \hat{\Phi}^\top c = \hat{K}_x^\top c = \sum_{i=1}^n c_i K(x, x_i).$$

Examples of Kernel: Linear Kernel

For $x, z \in \mathbb{R}^d$

$$K(x, z) = x^\top z$$

Proof

$$K(x, z) = \phi(x)^\top \phi(z)$$

with $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ defined as

$$\phi(x) = x.$$

Examples of Kernel: Affine Kernel

For $x, z \in \mathbb{R}^d$

$$K(x, z) = x^\top z + \alpha^2$$

Proof

$$K(x, z) = \phi(x)^\top \phi(z)$$

with $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d+1}$ defined as

$$\phi(x) = (x, \alpha).$$

Examples of Kernel: Polynomial Kernel of degree p

For $p \in \mathbb{N}$

$$K(x, z) = (xz + 1)^p \quad \text{with } x, z \in \mathbb{R}$$

Proof

$$(xz + 1)^p = \sum_{k=0}^p q_{p,k} (xz)^k = \phi(x)^\top \phi(z)$$

with $q_{p,k} = \frac{p!}{k!(p-k)!}$ and $\phi : \mathbb{R} \rightarrow \mathbb{R}^{p+1}$ defined as

$$\phi(x) = (\sqrt{q_{p,0}}, \sqrt{q_{p,1}}x, \sqrt{q_{p,2}}x^2, \dots, \sqrt{q_{p,k}}x^k, \dots, \sqrt{q_{p,p}}x^p)$$

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For $x, z \in \mathbb{R}^d$ it is defined as

$$K(x, z) = (x^\top z + 1)^p$$

Examples of Kernel: Polynomial Kernel of any degree

For $x, z \in [0, 1]$ and $0 < \alpha < 1$

$$K(x, z) = \frac{1}{1 - \alpha^2 xz}$$

Proof

$$\frac{1}{1 - \alpha^2 xz} = \sum_{k=0}^{\infty} (\alpha^2 xz)^k = \phi(x)^\top \phi(z)$$

with $\phi : \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}$ defined as

$$\phi(x) = (1, \alpha x, \alpha^2 x^2, \alpha^3 x^3, \dots)$$

Examples of Kernel: Polynomial Kernel of any degree

For $x, z \in [0, 1]$ and $0 < \alpha < 1$

$$K(x, z) = \frac{1}{1 - \alpha^2 xz}$$

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For $x, z \in \mathbb{R}^d$ it is defined as

$$K(x, z) = \frac{1}{1 - \alpha^2 x^\top z}$$

Examples of Kernel: Gaussian Kernel

For $X = \mathbb{R}$ and $\gamma > 0$ consider

$$K(x, x') = e^{-|x - \bar{x}|^2 \gamma}$$

Proof Let

$$\varphi_j(x) = x^{j-1} e^{-x^2 \gamma} \sqrt{\frac{(2\gamma)^{(j-1)}}{(j-1)!}}, \quad j = 2, \dots, \infty$$

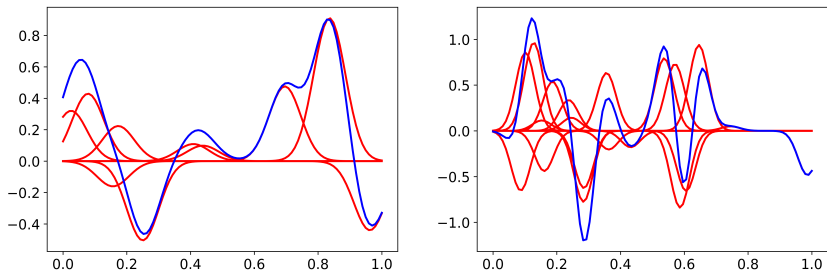
with $\varphi_1(x) = e^{-x^2 \gamma}$.

Then

$$\begin{aligned} \sum_{j=1}^{\infty} \varphi_j(x) \varphi_j(\bar{x}) &= \sum_{j=1}^{\infty} x^{j-1} e^{-x^2 \gamma} \sqrt{\frac{(2\gamma)^{j-1}}{(j-1)!}} \bar{x}^{j-1} e^{-\bar{x}^2 \gamma} \sqrt{\frac{(2\gamma)^{j-1}}{(j-1)!}} \\ &= e^{-x^2 \gamma} e^{-\bar{x}^2 \gamma} \sum_{j=1}^{\infty} \frac{(2\gamma)^{j-1}}{(j-1)!} (x\bar{x})^{j-1} = e^{-x^2 \gamma} e^{-\bar{x}^2 \gamma} e^{2x\bar{x}\gamma} \\ &= e^{-|x - \bar{x}|^2 \gamma} \end{aligned}$$

A key result

Functions defined by Gaussian kernels with large and small widths.



Kernel - Characterization

$K : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is a *Kernel* if it behaves like an inner product that is

1. it is symmetric

$$K(x, z) = K(z, x) \quad \text{for all } x, z \in \mathbb{R}^d$$

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$$K \text{ is p.d.} \quad \text{iff} \quad \hat{K} \text{ is p.d. for any } n \in \mathbb{N}, x_1, \dots, x_n \in \mathbb{R}^d$$

The first is easy to check, the second is quite difficult!

Kernel properties

Let $K_1 : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$, $K_2 : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$, $K_3 : \mathbb{R}^t \times \mathbb{R}^t \rightarrow \mathbb{R}$ be Kernels and $x, x' \in \mathbb{R}^d$, $z, z' \in \mathbb{R}^t$ and $\alpha, \beta > 0$ then the following are Kernels too

1. $\alpha K_1(x, x') + \beta K_2(x, x')$
2. $K_1(x, x')K_2(x, x')$
3. $p(K_1(x, x'))$ for any p a function whose polynomial expansion has only non-negative coefficients
4. $f(x)K_1(x, x')f(x')$ for any $f : \mathbb{R}^d \rightarrow \mathbb{R}$
5. $\frac{K_1(x, x')}{\sqrt{K_1(x, x)K_1(x', x')}}}$
6. $K_3(\psi(x), \psi(x'))$ for any $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^t$
7. $\alpha K_1(x, x') + \beta K_3(z, z')$
8. $K_1(x, x')K_3(z, z')$

Wrapping up

In this class we discussed how to deal with high dimensional non linear problems (feature maps and kernels). We also introduced the Representer Theorem.

Next class

... Neural Networks!